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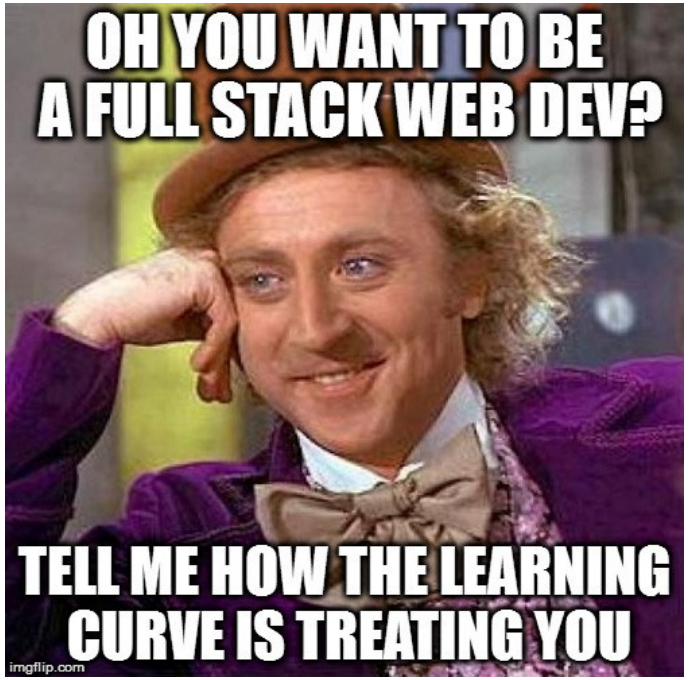
WEEK1

CN5006: Web and Mobile Application Development

Dr. Nadeem Qazi

Week1: Learning OutComes

- Module Structure and Assessment
- What is Web application
- How does Client & Server Communicate
- Web Application Components , Architecture & Protocols
- 3 Tier Architecture: Presentation ,Business and Data Layers
- Technological stacks: MERN
- Introduction to Node JS
- Node JS Installation, Architecture and Package Manger NPM
- Node.js Basics Modules
- Call-back & Promises



Module Contents & Assessments

Portfolio 15 Marks reflecting their learning experience on each of the weekly labs they performed. However, Every student must complete and attend the labs on their subscribed day otherwise the portfolio will not be accepted.	**Lab sessions in teaching weeks**
Mini Lab Assignments 10 Marks **The best of two would be considered.**	**Given in week 6,9to be submitted in week 9,12 respectively /**
Module Quizzes **(25 marks)**	**Two online quizzes of total 25 marks.** **1.On Line Quiz 1 week 7 (12 Marks covering Week 1 to Week6)** **2.On Line Quiz 2 week11 (13 Marks covering Week 7 to Week11)**
Coursework **50 Marks**	**Available of 5th Week. 26th Oct 2026 Submission date within 11th week time by 7th Dec 2026**
RESIT submission deadline	TBC

Coursework would be Group based, max 4 students in a Group

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12 Weeks Plan

Week 1: Introduction to Module, Web Arch MERN Introduction basic JS programming
Week 2: Introduction to MongoDB, Mongoose, Queries using a user interface on Local computer, optionally on Mongo DB atlas using MongoDB-compass
Week 3: JavaScript Revision , Express Framework, Middleware, REST API /
Week 4 : Introduction to React, JavaScript expressions in JSX props, functions,
Week 5: Functional Components, Event handling, state management using Hook + Coursework Description
Week 6 : Controlled Form using HOOK. First Assignment discussion 10 Marks submission deadline
Week 7: React Routing, React Form for SPA.On-Line Quiz 1 (12 Marks,Week 1 to Week6)
Week 8: Connecting React to Databases
Week 9: Version control system, Revision Q/A from week 1 to 8. Second Assignment discussion 10
Week10: Class Components
Week11: Advanced Topic in React Mobile/Android application etc. React Management Tools. React Context API vs React Redux
Week 12: Revision

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Entry-Level Job Opportunities

- Junior Full-Stack Developer
- Junior React Developer
- Web Developer
- JavaScript Developer
- Graduate Software Engineer
- API Developer
- Technical Consultant



Advanced Career Paths

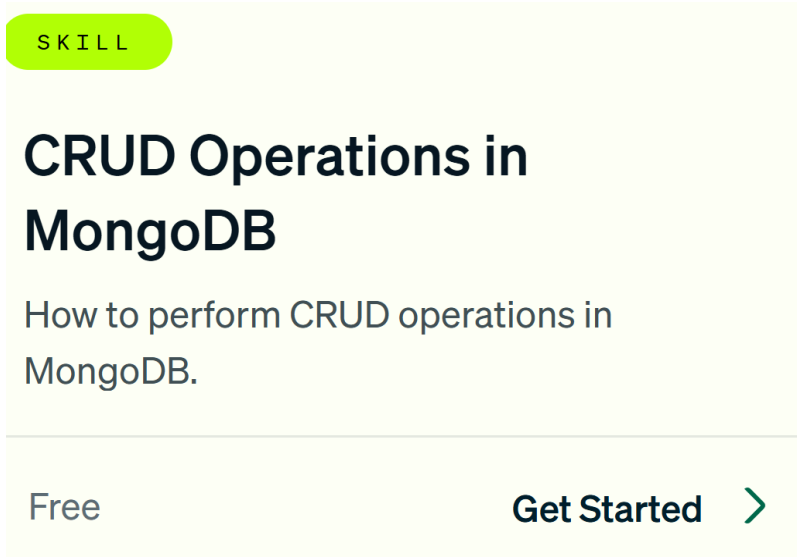
MERN Stack Developer

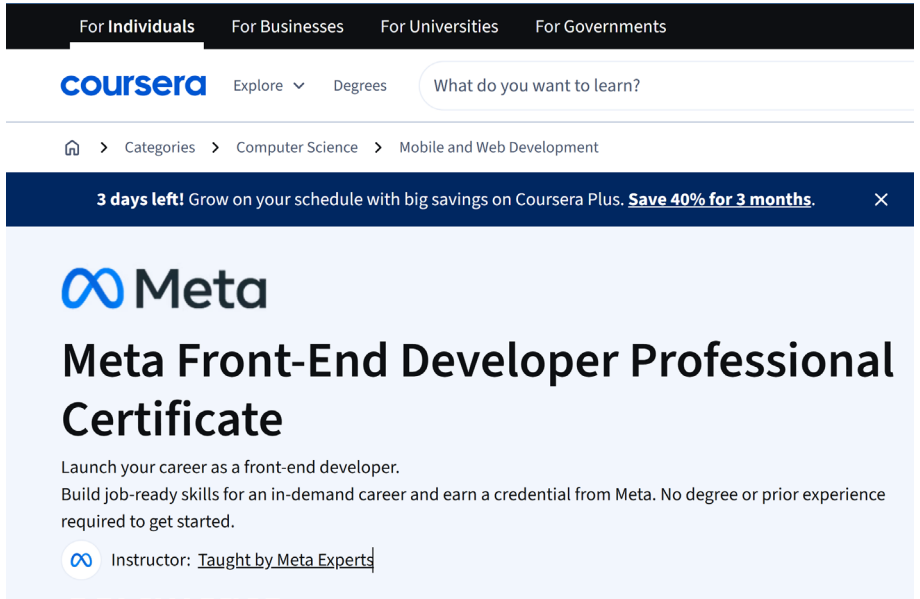
- Front-End Engineer
- Backend Node.js Developer
- Cloud Application Developer
- DevOps Engineer (with cloud certifications)
- React Native Mobile Developer
- AI Application Developer

```
Student
 ↓
Junior Developer
 ↓
Software Engineer
 ↓
Senior Engineer
 ↓
Solution Architect / Technical Lead
```

Recommended Certifications

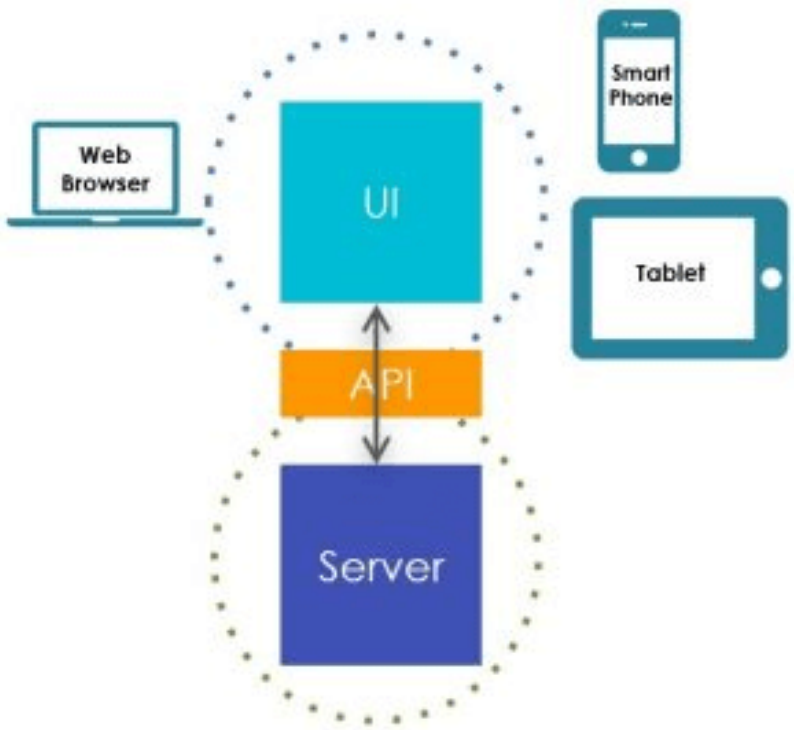
MongoDB Associate Developer – learn.mongodb.com
FreeCodeCamp JavaScript Certification – freecodecamp.org
Meta Front-End Developer – coursera.org
Meta Back-End Developer – coursera.org
Git & GitHub – coursera.org





What is Web application

- A web application is computer software consisting of Web pages with a customised user interface run through a web browser.
- Web architecture is the design and structure of components that work together to deliver a web application.
- https://www.youtube.com/watch?v=cDmAQm8X5XA

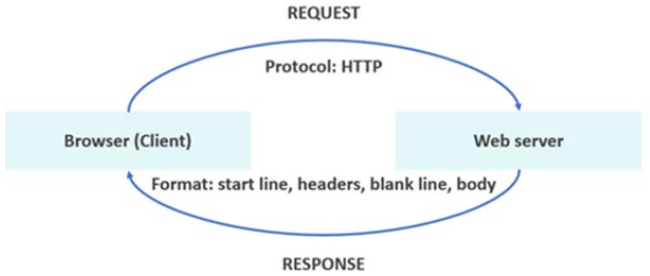


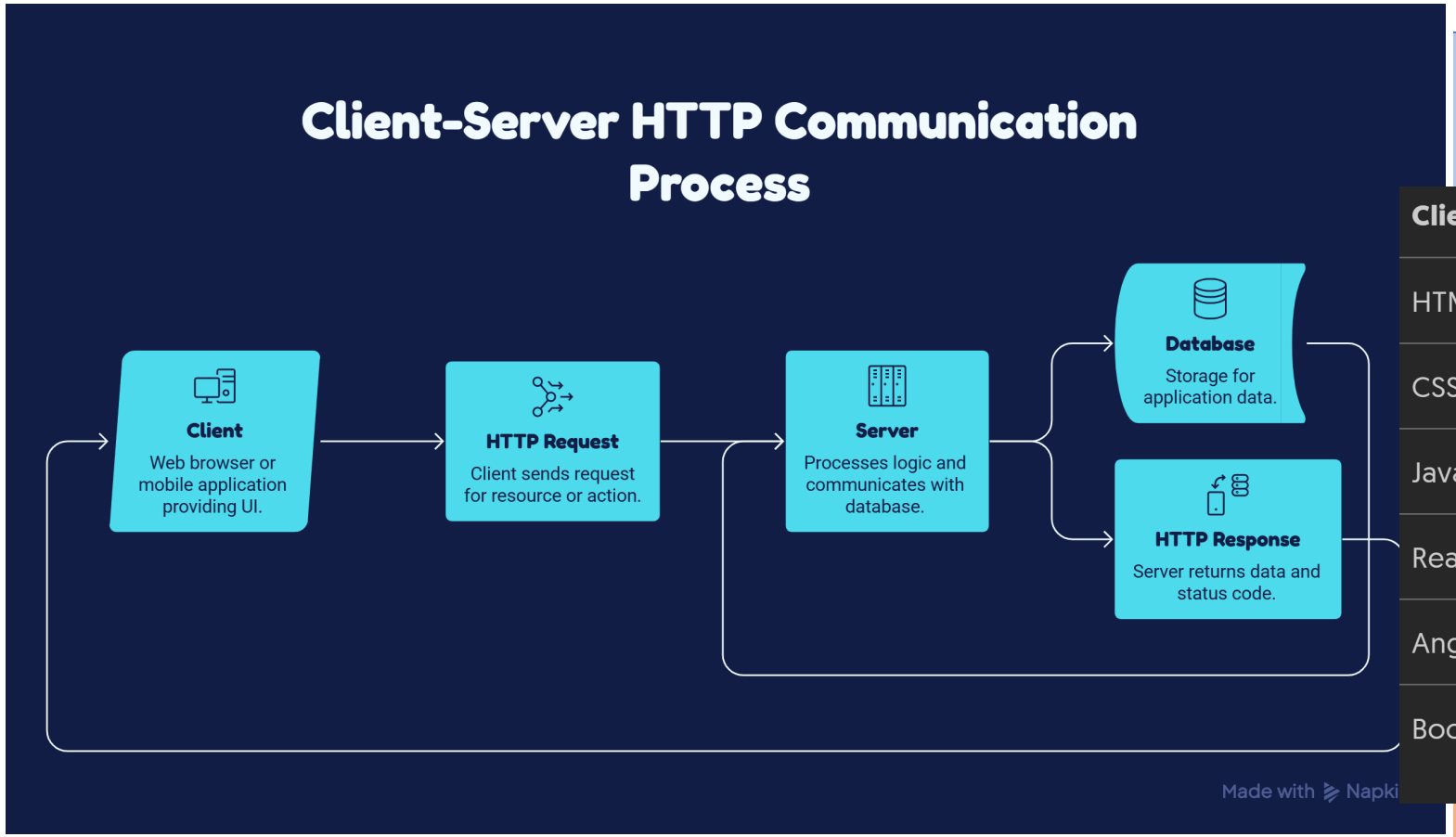
Examples of the projects based on Module

- Student Management System
- E-Commerce Website
- Online Appointment Booking System
- Inventory System
- AI Chatbot Interface
- Event Management Platform

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How a Web Application Works





Client Starts a Request through

1. The URL for the resource
2. A method, such as:
   i. GET (for fetching data from the server) or

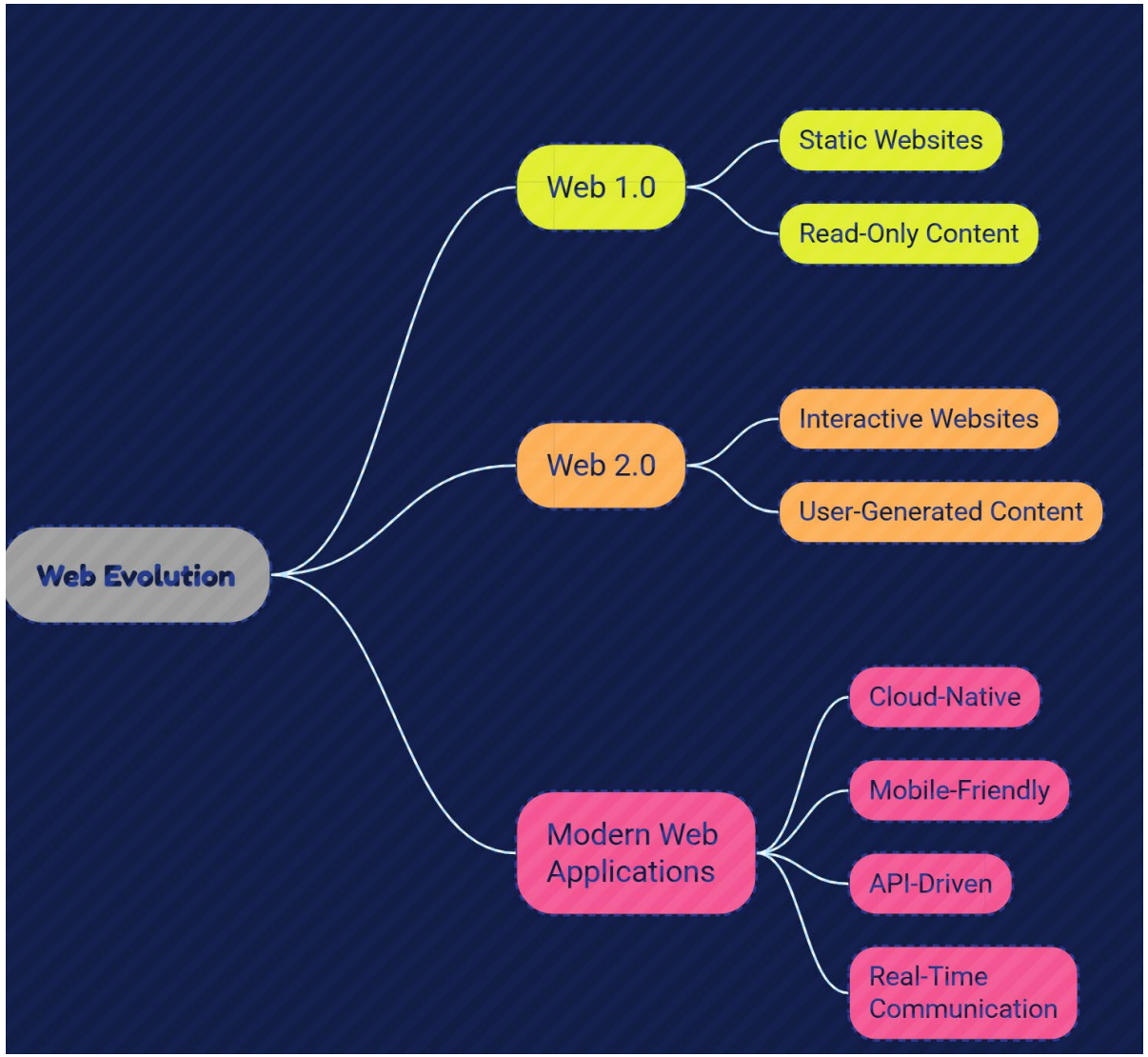
Client Side (Frontend)	Server Side (Backend)
HTML	Node.js
CSS	Python
JavaScript	Java
React	.NET
Angular / Vue	PHP
Bootstrap / Tailwind	Databases (MySQL, PostgreSQL, MongoDB)

2. A list of headers (response, request headers carrying information such as message size, content type, etc.)
3. The body which contains the information

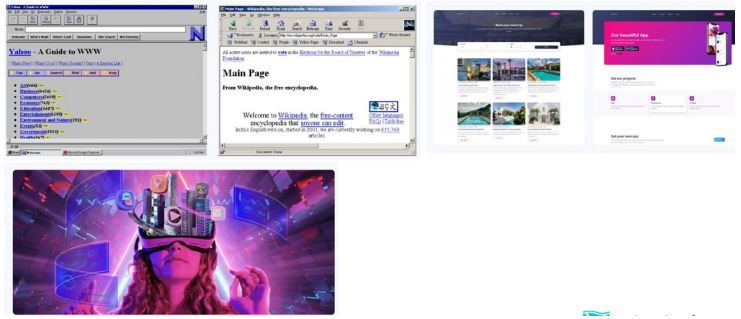
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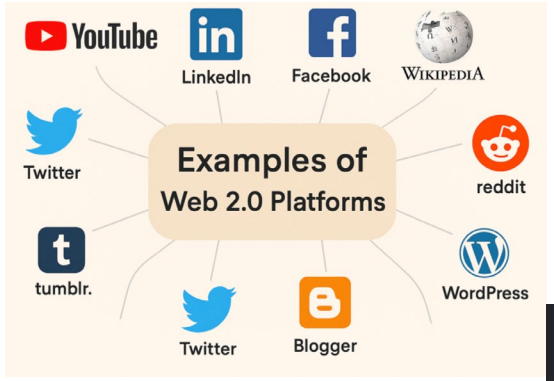
Web Architecture

- Overview of web 1.0 ,2.0 and web 3.0 Architecture



Web 1.0



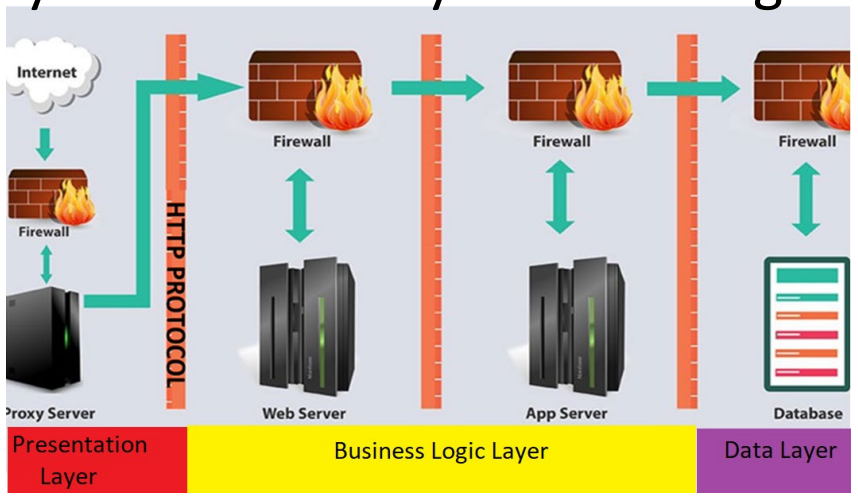


Web App Example	Core UX Focus	Key Technology Under the Hood
Google Docs	Real-time multi-user synchronization	Operational Transformation (OT) & WebSockets
Netflix	Media rendering & personalization at scale	Adaptive Bitrate Streaming & AI Recommend Engines
Gmail	Single-page fluid navigation	Asynchronous JavaScript (AJAX) background fetching
Airbnb	Dynamic transactional marketplace logic	Microservices architecture & interactive mapping APIs

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3-Tier Application: Presentation Layer

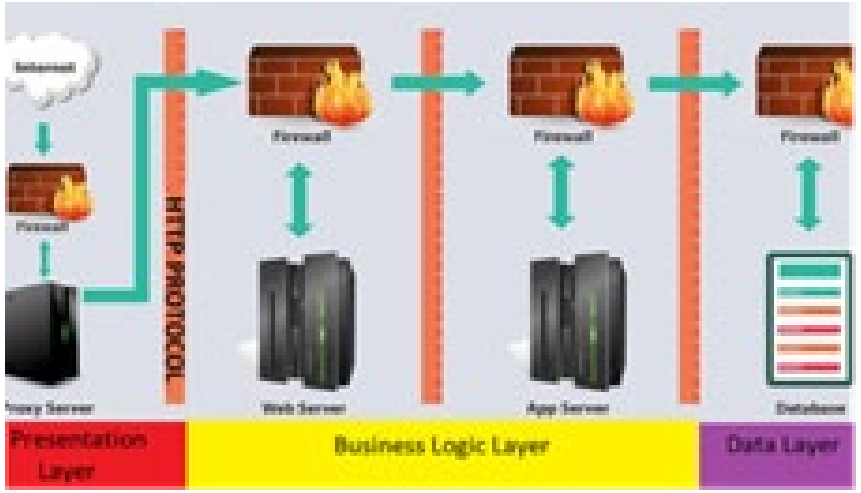
- An 3-Tier Application is distributed among three separate computers/layers in a distributed network, however can be on the same system.
- Presentation/client layer that deals with user interface of the application and that communicates with other tiers through API.
- This layer contains the design, content structure, animation, behaviour, and everything that you see on the screen while you open a web application, website or mobile app.
- Technologies like HTML, CSS and JavaScript are used to render the content for the user.
- It calls functions of other layers so that they can be integrated with each other.



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3 –Tier Architecture :Business Layer

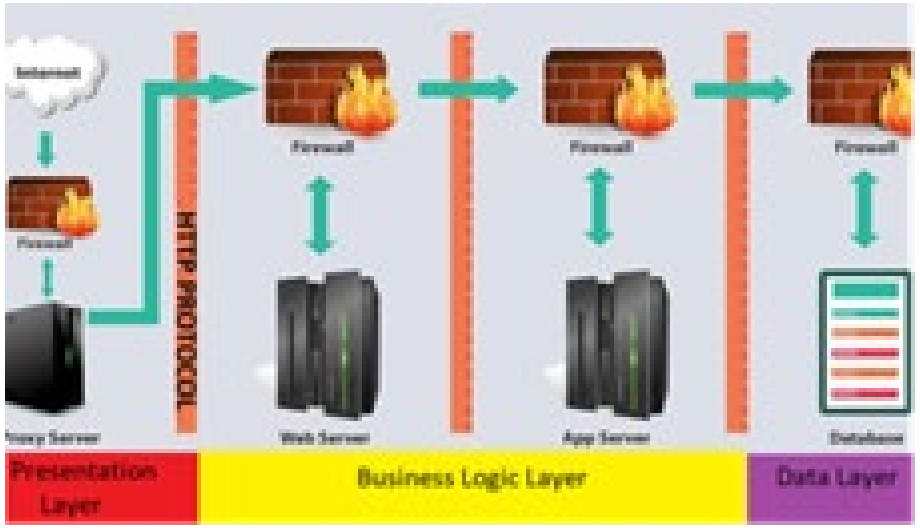
- Business Layer: Acting as interface b/w presentation and data layer, accepts the data from the presentation layer and passes it to the data layer, which stores the data in the relational database.
- All business logic including validation of data, calculations, data insertion/ modification are written under business logic layer.
- It is usually implemented these days using technologies like Ruby, Python, PHP, Java, C++ or ASP.NET.
- This layer is concerned more about the data, the data validation, the dynamic content processing and generating the content to be delivered to the user.



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3 –Tier Architecture ,Data Layer

- Data Layer
- This layer provides functionality to store and interact with the data, typically in the form of a database and access this data through an API.
- This functions defined in this receives the data from the business layer and performs the necessary operation into the database.



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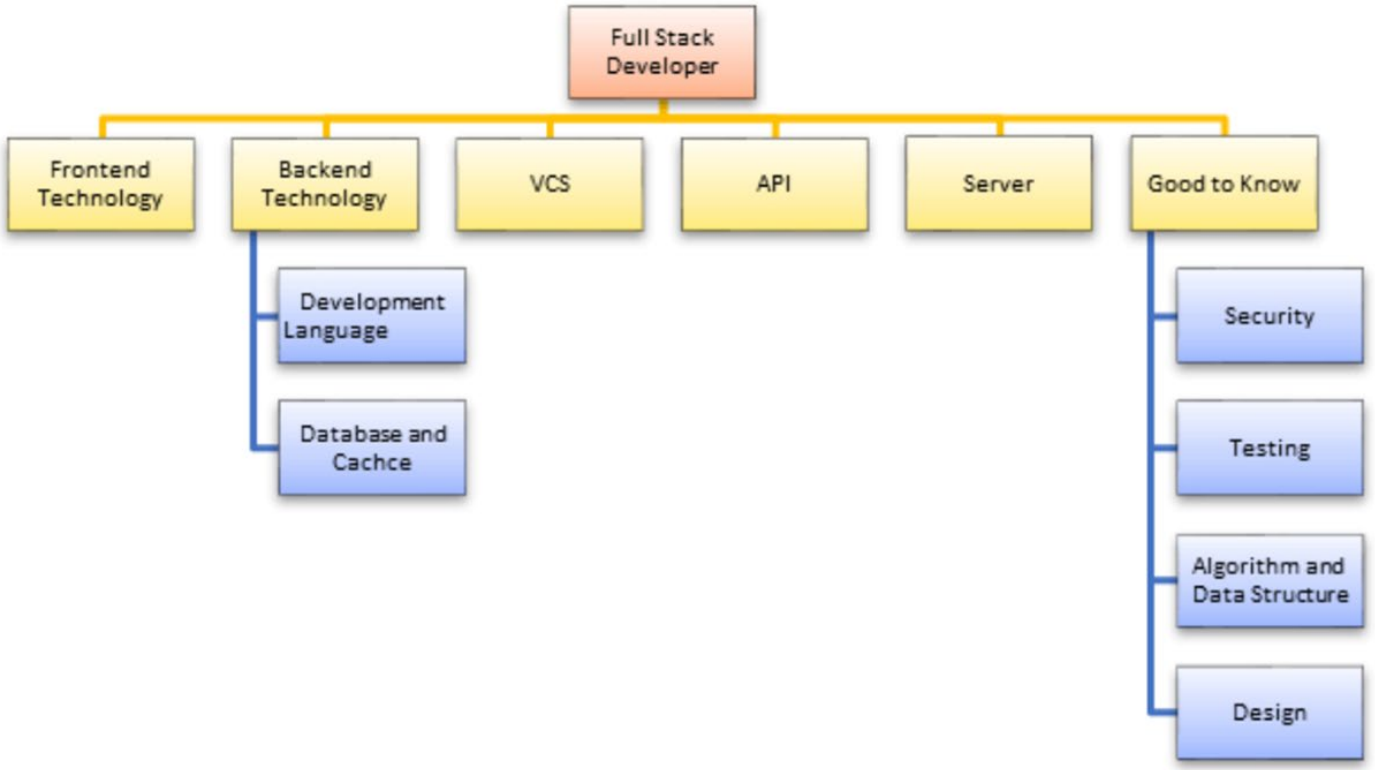
Modern Application Development

- A modern application expect to have an undefined number of clients
  - the client is a UI using the React JavaScript library,
  - A downstream application that connects to the application through an API.
  - Version Control for Software
- Data is available in a generic, consumable format, such as JSON.
- A modern application provides an RESTful APIs or GraphQL over HTTP(S) for accessing data and services, available through the GUI or CLI.
- Modern applications are built on top of a modern stack , the popular version are based on Java, Python, Node, Ruby, PHP, React etc.

Web App Example	Core UX Focus	Key Technology Under the Hood
Google Docs	Real-time multi-user synchronization	Operational Transformation (OT) & WebSockets
Netflix	Media rendering & personalization at scale	Adaptive Bitrate Streaming & AI Recommend Engines
Gmail	Single-page fluid navigation	Asynchronous JavaScript (AJAX) background fetching
Airbnb	Dynamic transactional marketplace logic	Microservices architecture & interactive mapping APIs

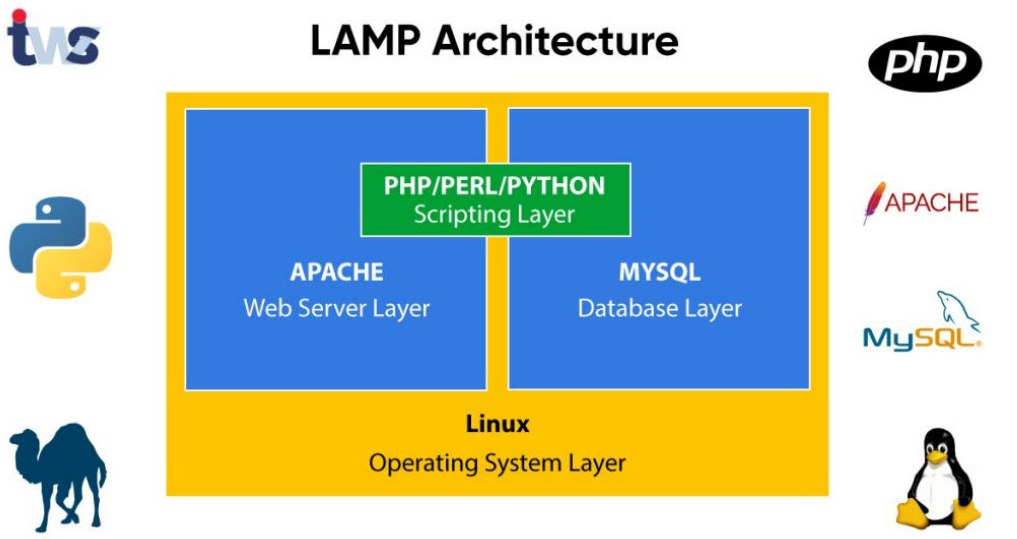
What Is a Stack?

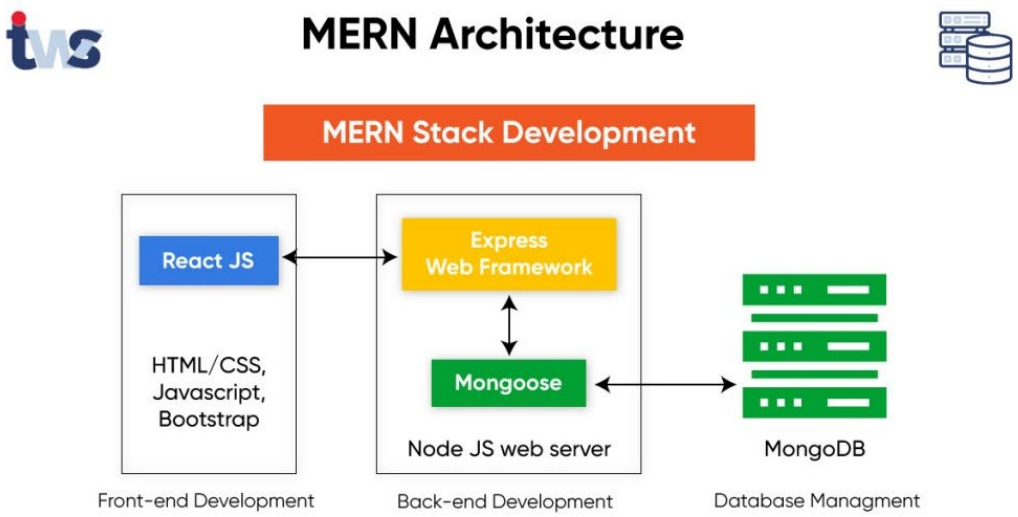
- A “Stack” is a combination of programming languages, technologies or even software products. Development stacks are further divided into two categories – Technology stack & application stack.
- developers who can work on almost all the aspects of a mobile/web app development including work on both – Frontend & Backend. project are called full-stack developers

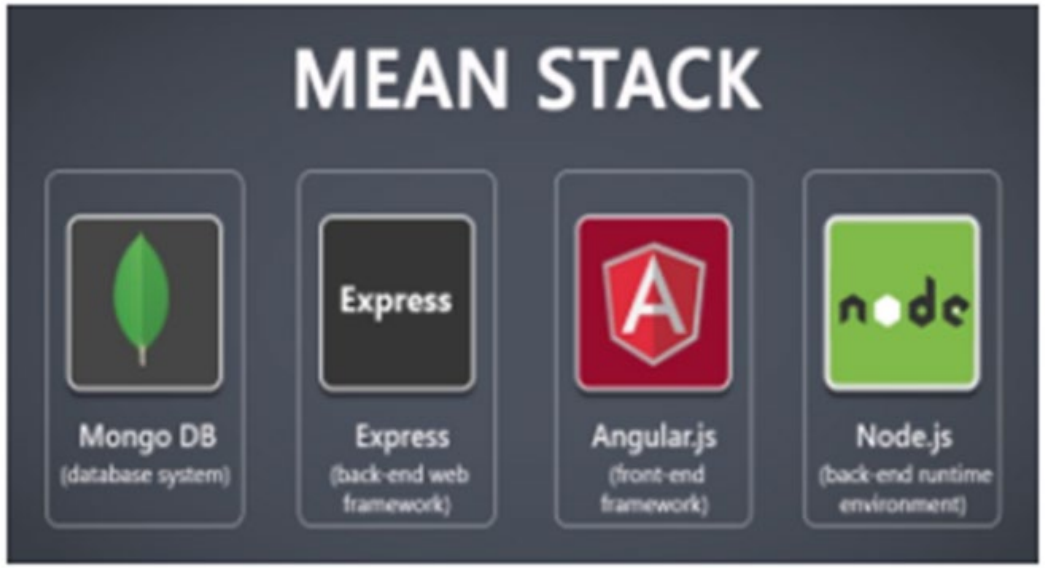


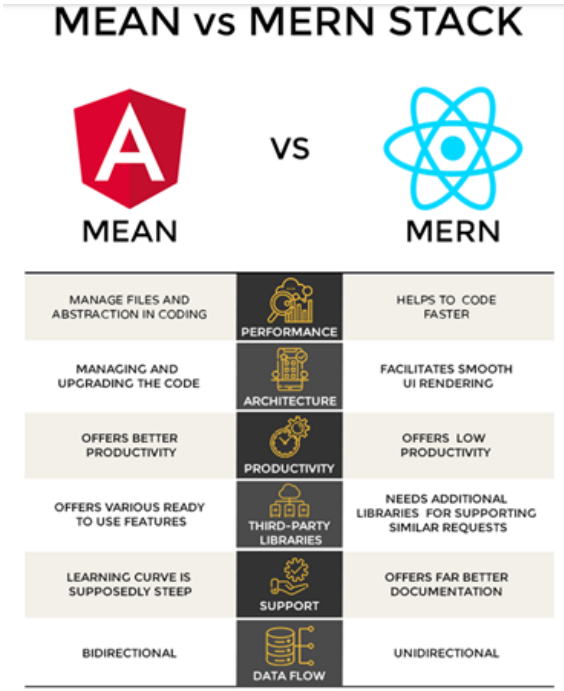
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Technological Stacks



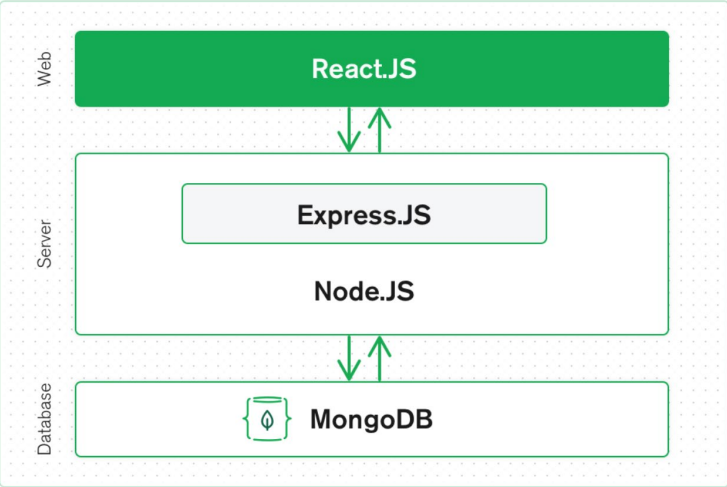


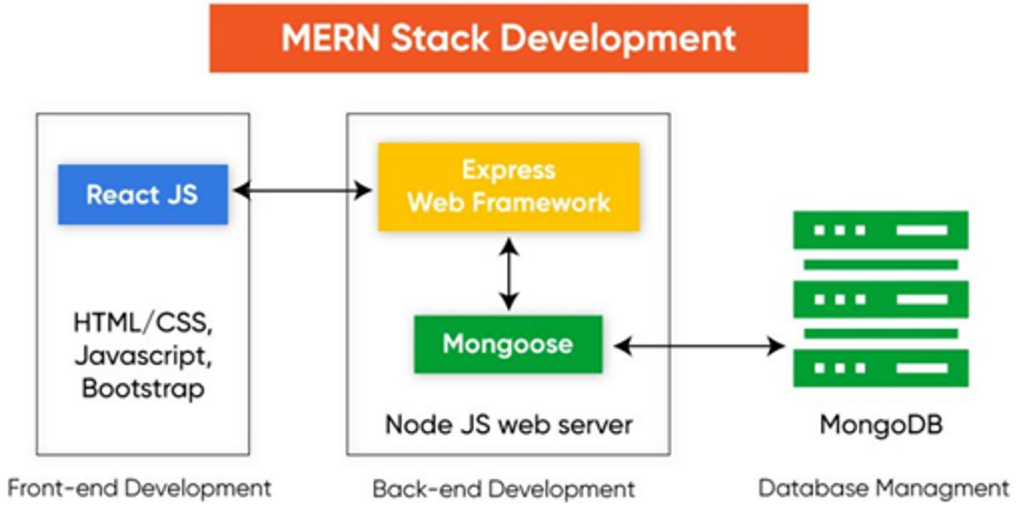




MERN Stack

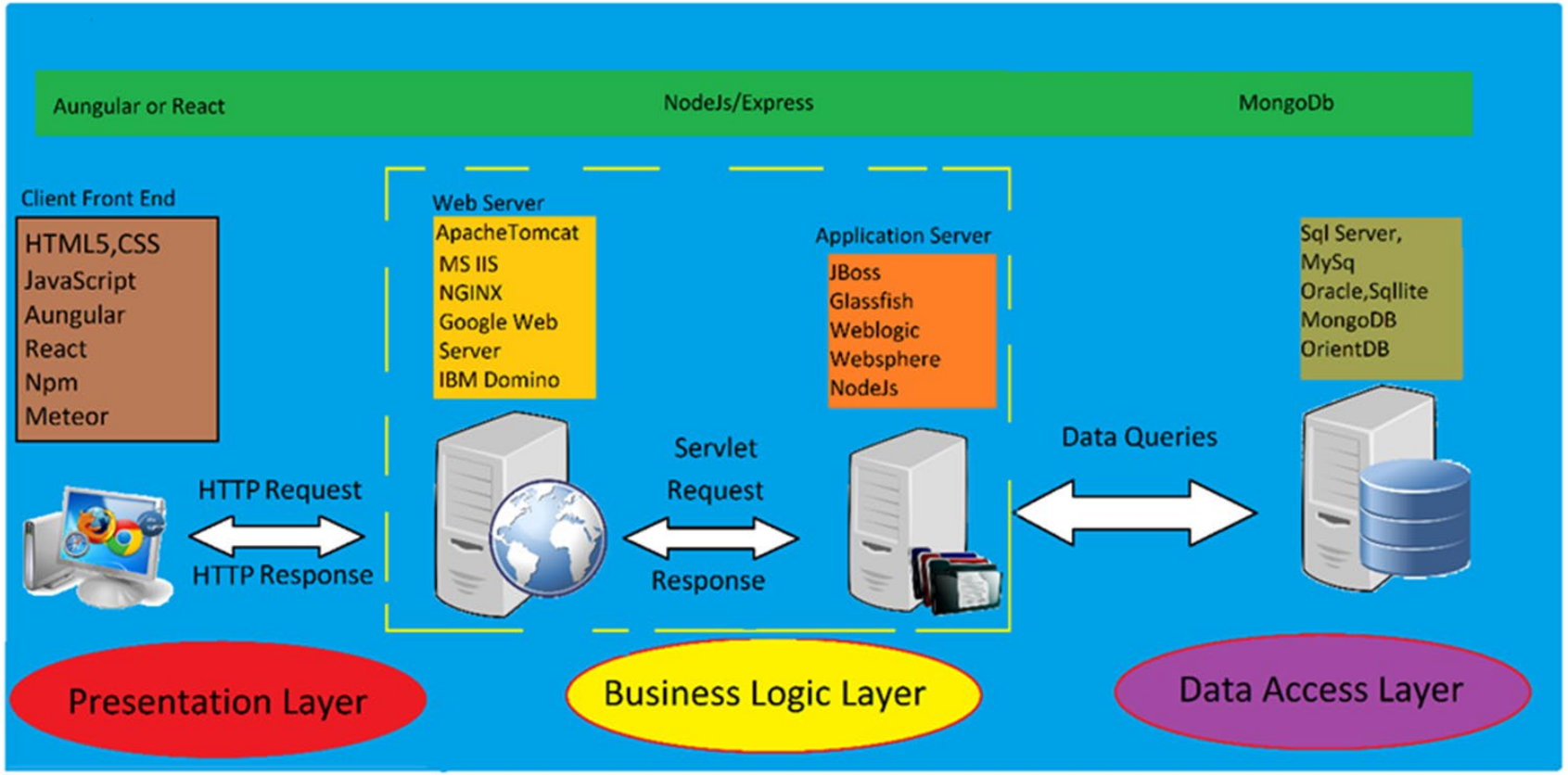
- MERN is the acronym for *M*ongoDB, *E*xpress JS, *R*eact JS and *N*ode JS. The MERN stack is a combination of the these technologies, all based on JavaScript, used to build advanced web applications.
- Full stack development from front-end development to back-end development using JavaScript and JSON.
- Node.js : Back end JavaScript runtime environment used to implement Business Logic Layer.
  - Express: A fast web framework for Node.js.
- React : Java Script library for building client side or Front end development
- MongoDB: Document database for storing data as JSON document at back end.





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MERN 3 Tier Applications Tools and components



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How AI Uses MERN Technologies

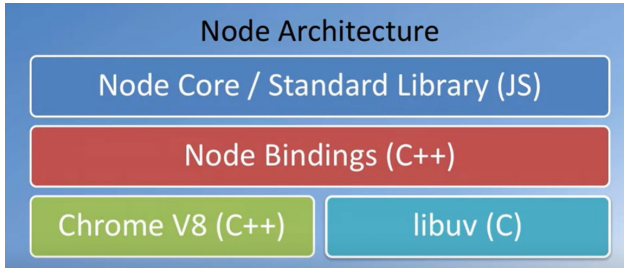
- ChatGPT web applications
- AI dashboards
- AI APIs
- Agentic AI interfaces
- Cybersecurity platforms

Technology	Purpose
React	User Interface
Express	Routing
Node.js	Server
MongoDB	Database

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What is NodeJS

https://www.youtube.com/watch?v=uVwtVBpw7RQ



1. Node is a Server side JavaScript runtime environment built on Chrome's V8 JavaScript engine.
2. Allows developers to make different servers like DNS server, TCP server, HTTP server, and more through built in APIs
3. Node has an event-driven architecture, allows building scalable light weight real time web applications
4. Its unique asynchronous, non-blocking I/O model eliminates the waiting approach to serving requests.
5. Node's package manager npm, is the largest open source ecosystem in the world
6. **Apps Types: Single page Realtime DataStreaming, RestAPI Server side Rendered CL tools & HW Programming**

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Lab Work

- Javascript Program to Check if a number is Positive, Negative, or Zero
- Write the following program in the index.js

```
// app.js or index.js
const prompt = require('prompt-sync')(); // This line is essential
console.log("starting")
const name = prompt('Enter your name: ');
console.log(`Hello, ${name}`);
// program that checks if the number is positive, negative or zero
// input from the user
const number = parseInt(prompt("Enter a number: "));

// check if number is greater than 0
if (number > 0) {
    console.log("The number is positive");
}

// check if number is 0
else if (number == 0) {
  console.log("The number is zero");
}

// if number is less than 0
else {
    console.log("The number is negative");
}
```


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Node Package Manager

- The Node Package Manager is the manager for the Node ecosystem and automatically gets installed during the installation of the Nodes
- It manages all the Node modules and packages that have been made publicly available by many different users.
- When using npm, you're most likely going to be using the command line tool for the majority of your interactions.
- A typical Node package consists of JavaScript files together with a file called package.json which is the manifest file for this node module.
- For better understanding please watch another video on NPM.

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Some useful NPM packages

React

React is a JavaScript library for building user interfaces.

express

a de-facto web development framework for Node.js,

mongodb & mongoos

MongoDB wrappers to provide the API for MongoDB object databases in Node.js.

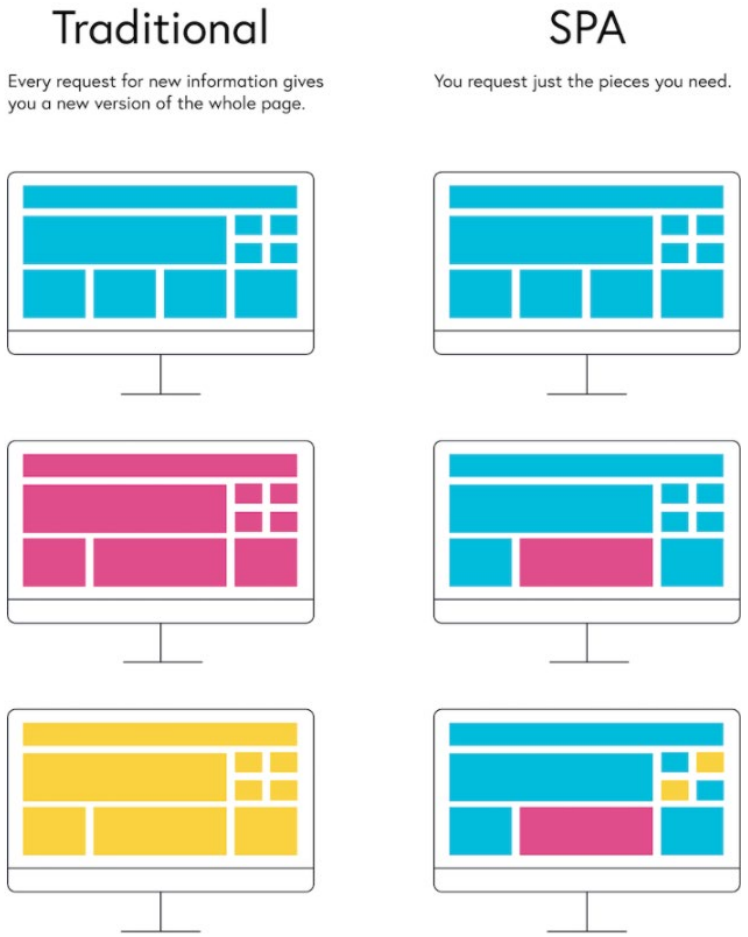
react-dom

React package for working with the DOM.

For Installation please follow the guide MERNINSTALLATION.PDF

Node JS Application Types

- Single-page apps (SPAs): The single page application is a web application or website that interacts with the user by dynamically rewriting the current page, rather than loading entire new pages from the server.
- SPA are different than a traditional page load where the server re-renders a full page with every click you make and sends it to your browser.
- The SPA only sends what you need with each click, and your browser renders that information.
- SPAs include social networking apps, email or map apps, online text or drawing tools, etc.



Node JS Application Type

- Real-time apps (RTAs): Web apps that enable users to receive information as soon as it's published by an author example include instant messaging apps or chat rooms, video conference apps, etc.
- Data streaming apps: These are apps that send data/content as it arrives while keeping the connection open to continue downloading further content. video- & audio-streaming apps.
- REST APIs: Are interfaces that provide data for someone other app. For example, a Facebook API for Posts.
- Server-side rendered apps (SSRs): These web apps can run on both the client (front-end) and the server (back-end) allowing pages that are dynamic to display HTML. SSRs utilize SPA methods in that they don't need to reload every time you use it.
- Command line tools: These allow you to automate repetitive tasks and then distribute your tool across the vast Node.js ecosystem. An example of a command line tool is cURL, which stand for client URL and is used to download content from an internet URL.
- NPM contains more than 80 packages for Arduino controllers, raspberry pi,

  Intel IoT Edison, various sensors, and Bluetooth devices.

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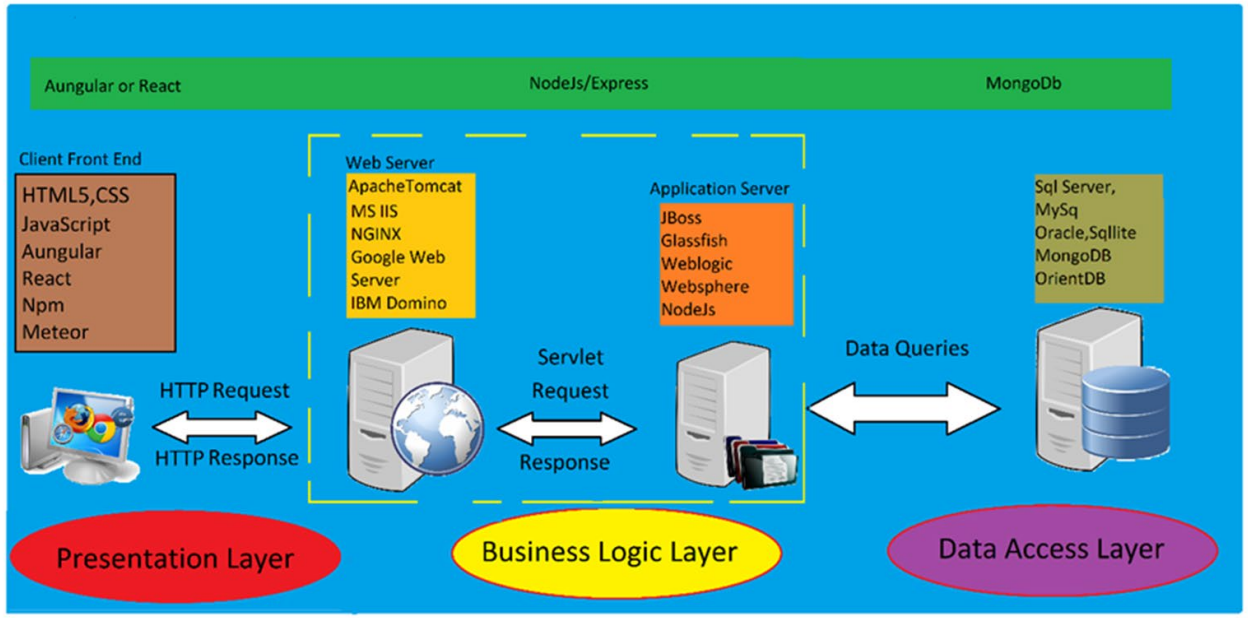
Quick Recap

- ✓Web Application Architecture
- ✓HTTP Protocol and methods
- ✓N-Tier Application :Presentation, Business and Data Layer
- ✓Web Component
- ✓Modern Application Development
- ✓MERN Stack
- ✓Nodes Modules
- ✓ JS Arrow Functions

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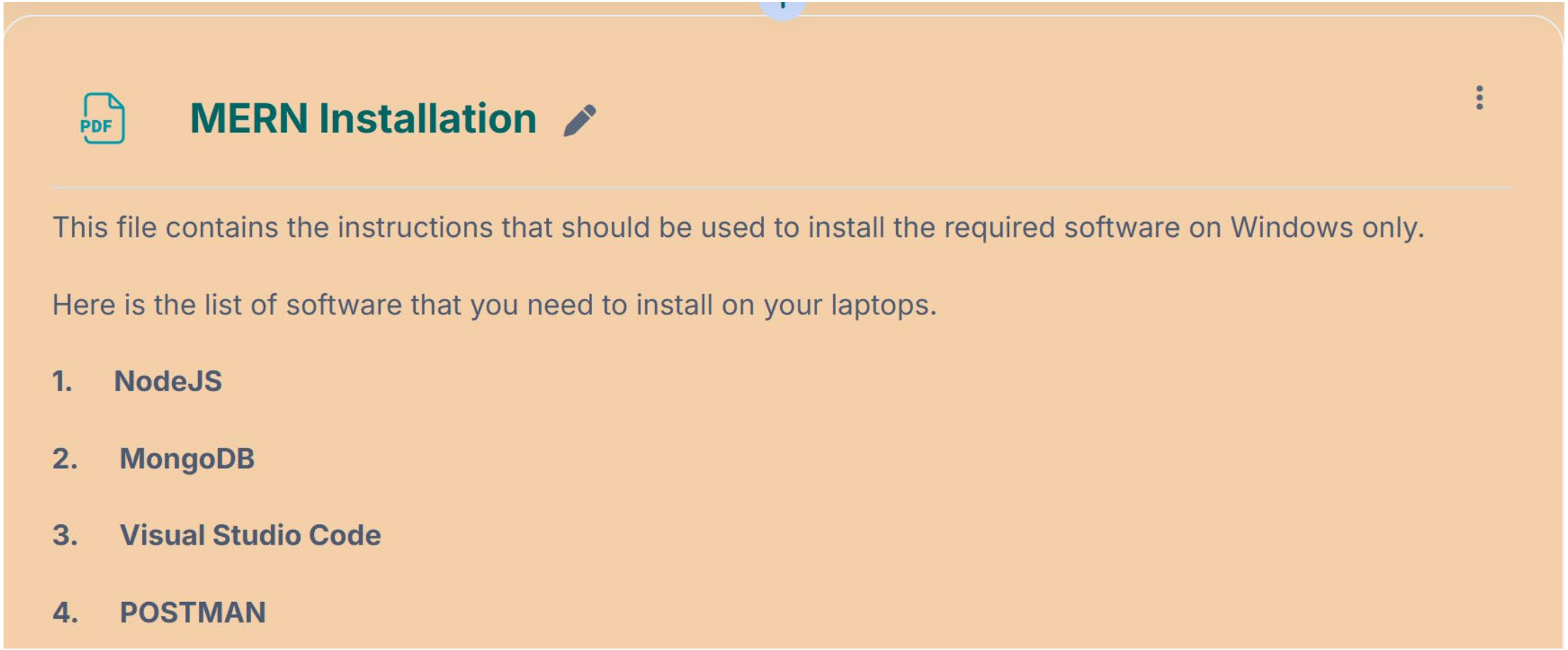
MERN Installation

- Follow the instruction as mentioned in the MERN installation.pdf
- https://moodle.uel.ac.uk/mod/resource/view.php?id=2996012
- Week 2 MongoDB
-



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MernInstallation.pdf



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Lab Work

Write the following program in the index.js

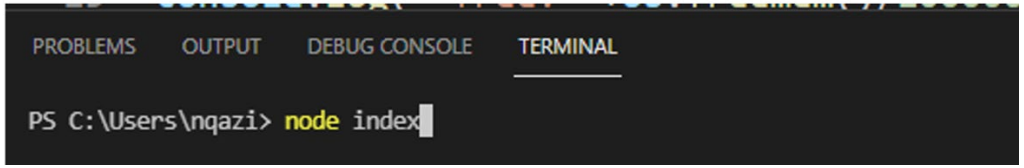
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// input from the user
const number = parseInt(prompt("Enter a number: "));

// check if number is greater than 0
if (number > 0) {
    console.log("The number is positive");
}

// check if number is 0
else if (number == 0) {
  console.log("The number is zero");
}

// if number is less than 0
else {
     console.log("The number is negative");
}
```

In the terminal window type node index and hit enter.



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How do u compile a programme in Visual Studio Code. node filename

Now Star writing the code

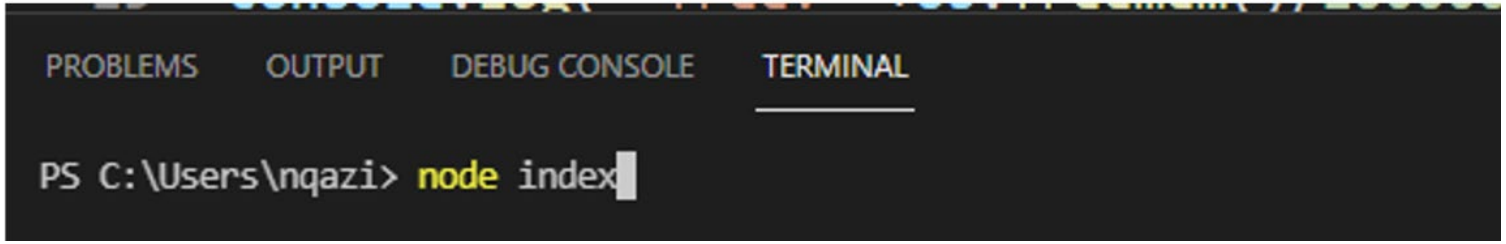
Exercise 1:

Write a code in Java script (index.js)

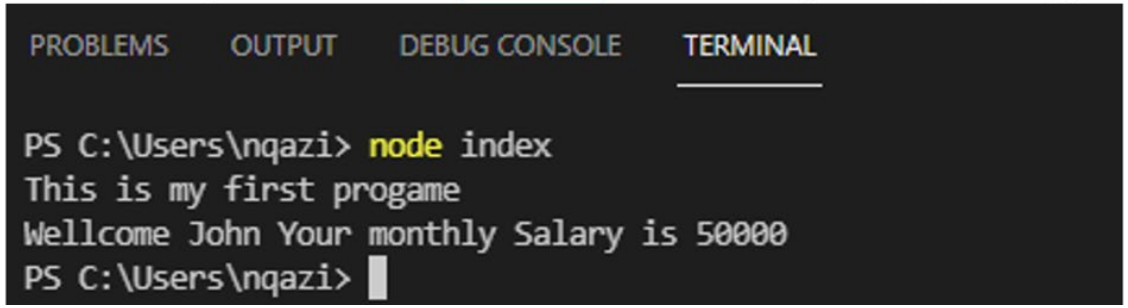
Write down following line

```
console.log("This is my first programe ")
console.log("Well come John your month salary is 500000")
```

In the terminal window type node index and hit enter.



You will see you code will be executed and following message will appear in the Terminal window.



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Lab Assignment 1:

Write a JS programme using VSC , run it in terminal and submit the output to secure 2 marks. Here is the code you can use.

- const num1 = 5;
- const num2 = 3;

- // add two numbers
- const sum = num1 + num2;

- // display the sum
- console.log('The sum of ' + num1 + ' and ' + num2 + ' is: ' + sum);

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Lab 1 Portfolio

Portfolio Exercise

Task Description:
Create a JavaScript program that performs basic arithmetic operations—addition, subtraction, multiplication, and division—on two or more numbers. Include this program in your **Week 1 portfolio**.

Reflection:
Write a brief reflection on this week's lab activities. Discuss your learning experience, any challenges you faced, and key insights gained. Add this reflection to your portfolio as well.

• Please remember that your portfolio is due in Week 11 and must be submitted on Moodle, However, ensure that you have shown your Lab instructor the completed Lab 1 Week exercise beforehand.

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